DS Capstone Check in 2

Team 1

Scraping Update

- Extraction
  - Done via Arxiv python package and put into pandas dataframe
  - Split separate sections of paper with regular expressions
    - Also includes cues for both hypothesis and conclusion
  - Eventually only used cues for just hypothesis
    - Inconsistency with extraction for both sections simultaneously
  - Dropped papers with empty hypothesis or conclusion
- Paper Subjects
  - “arxiv_extracted_final” - Artificial Intelligence
  - .csv files with other keywords to test functionality of embedding pipelines

Pre Processing Updates

Timeline

- There is an obvious need for Data Pre-Processing
    - missing sections, poorly extracted text, outliers, inconsistent hypothesis definitions, excessively long sections, PDF artifacts, and unreliable hypothesis-conclusion pairs.
- Goal: (X_Hypothesis, Y_Conclusion)
- Pipeline Development
    - PreProcessing File in Git, first python script
- Development of the Hypothesis Fallback Extractor “--extract-hypothesis-if-missing”
- Results

Embedding Updates

Embedded Diff Vectors

- CalculateBGEEmbeddedDifferents.py
- CalculateQwenEmbeddedDifferents.py

Embedded Single Vectors

- CalculateQwenSingleEmbeddings.py

Configuration Cell
Set Pipeline Dataset paths and Script parameters

Data Loading Cell: Loads the initial *Raw* dataset into a Panda in notebook memory

Pre-processing Cell

pp.py

CalculateBGEEmbeddedDifferents.py

CalculateQwenEmbeddedDifferents.py

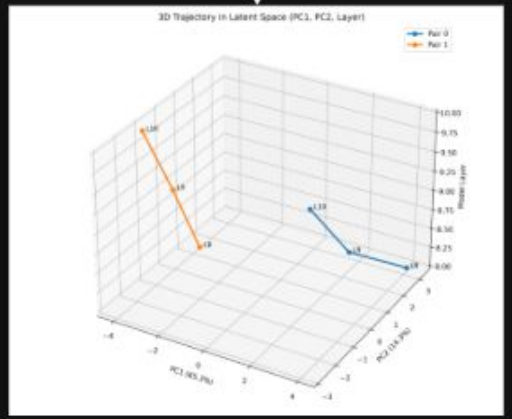
CalculateQwenSingleEmbeddings.py

Embedding Cell

Visualization

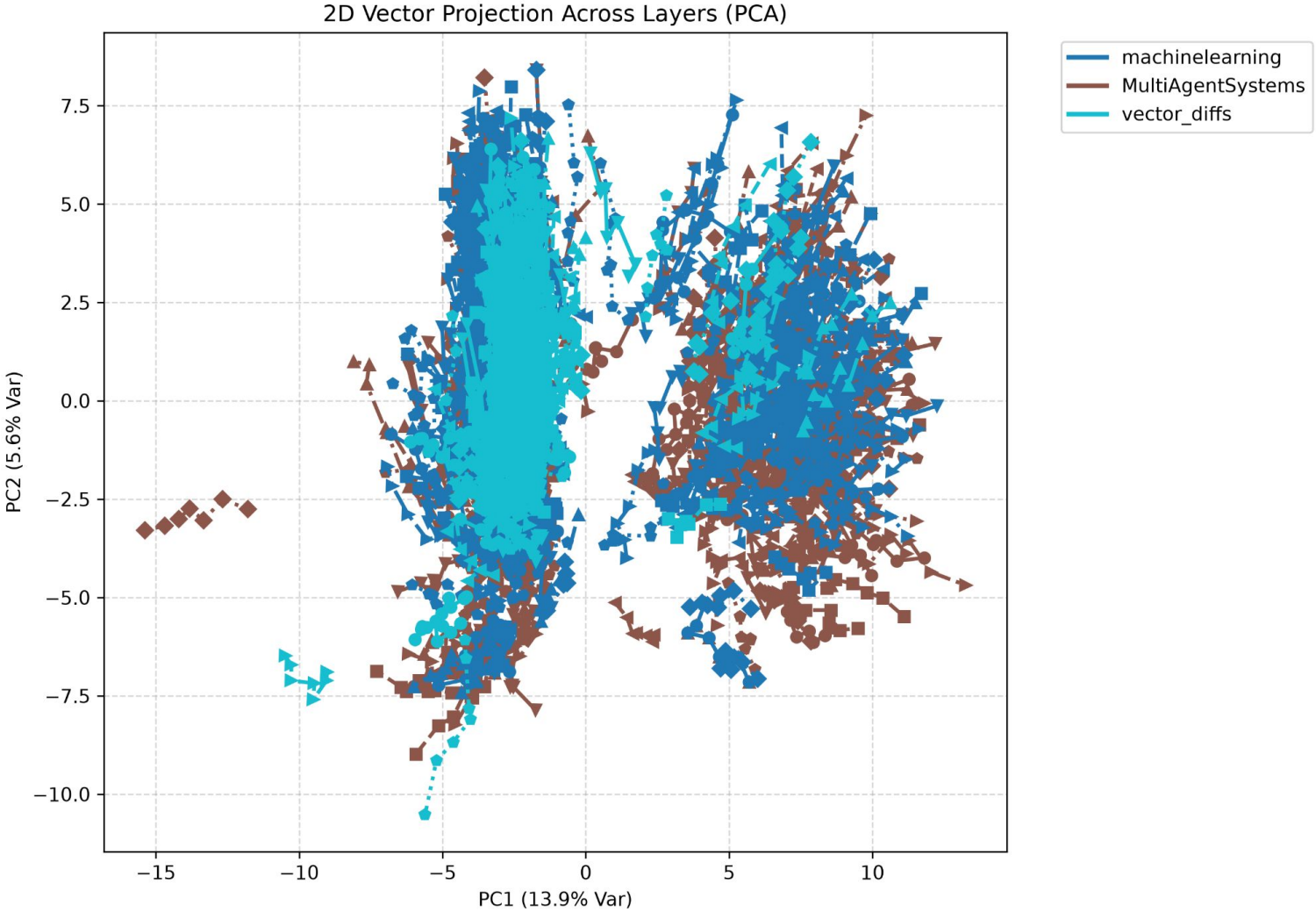
umap.py

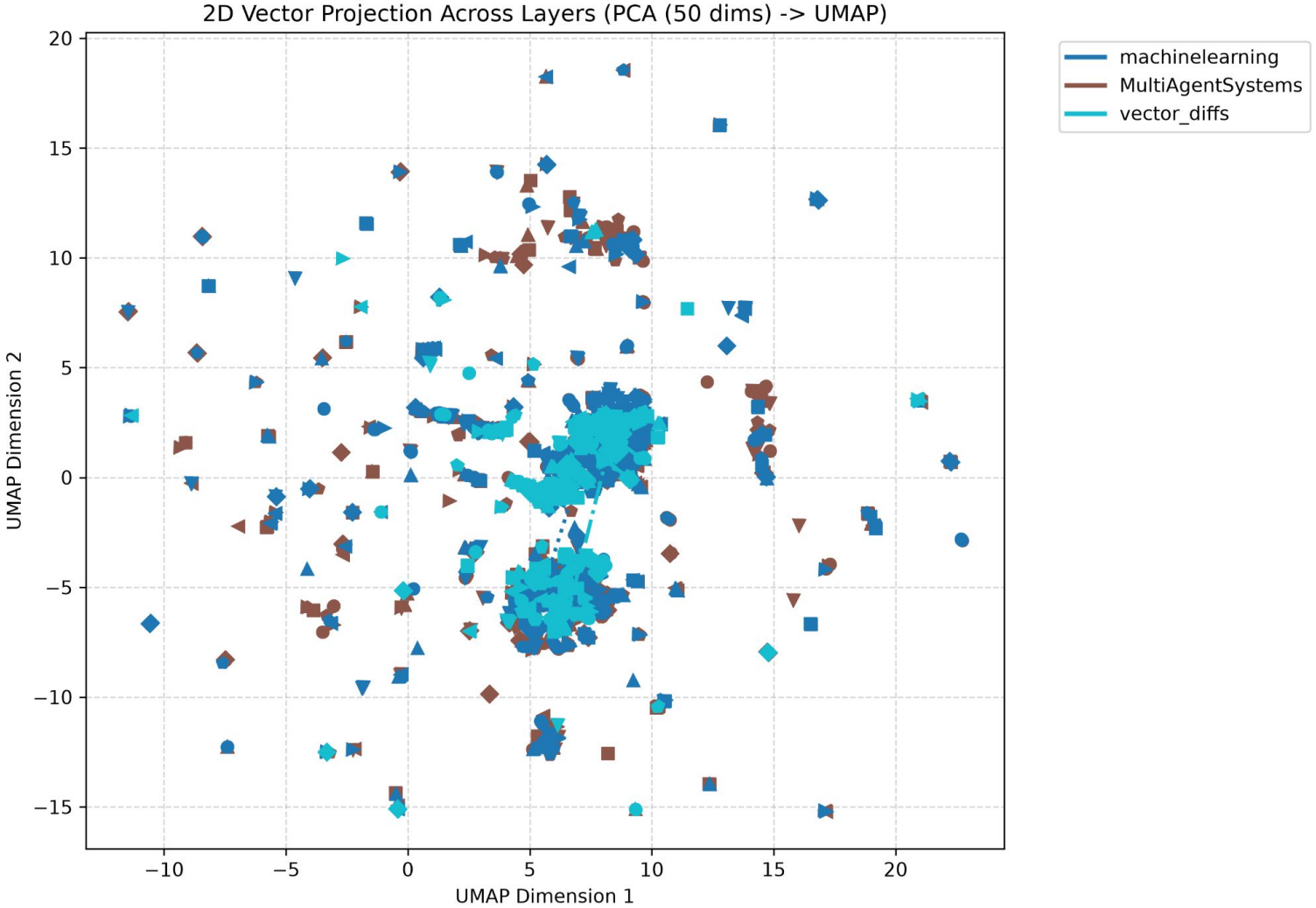
pca.py

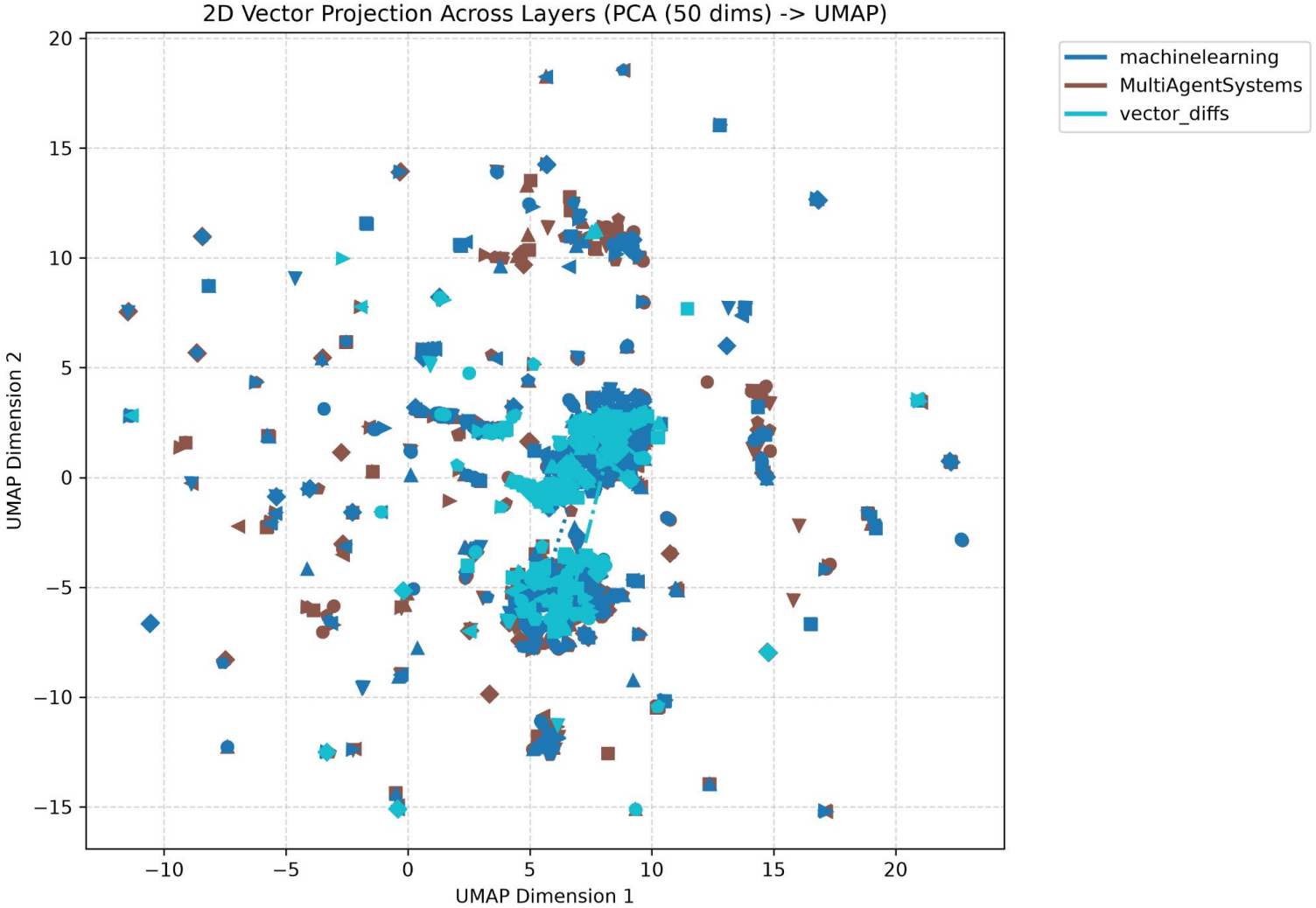


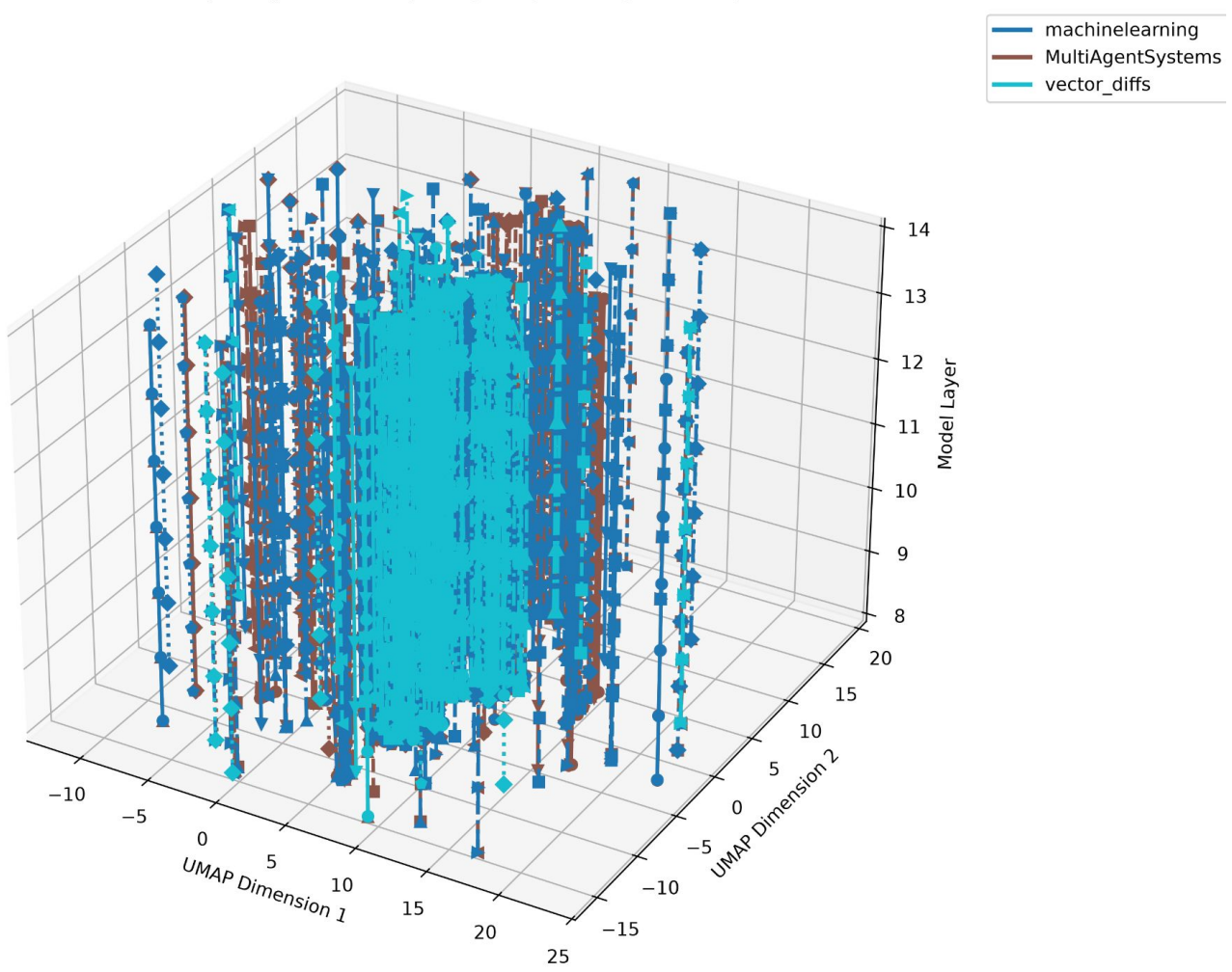
Visualization Updates

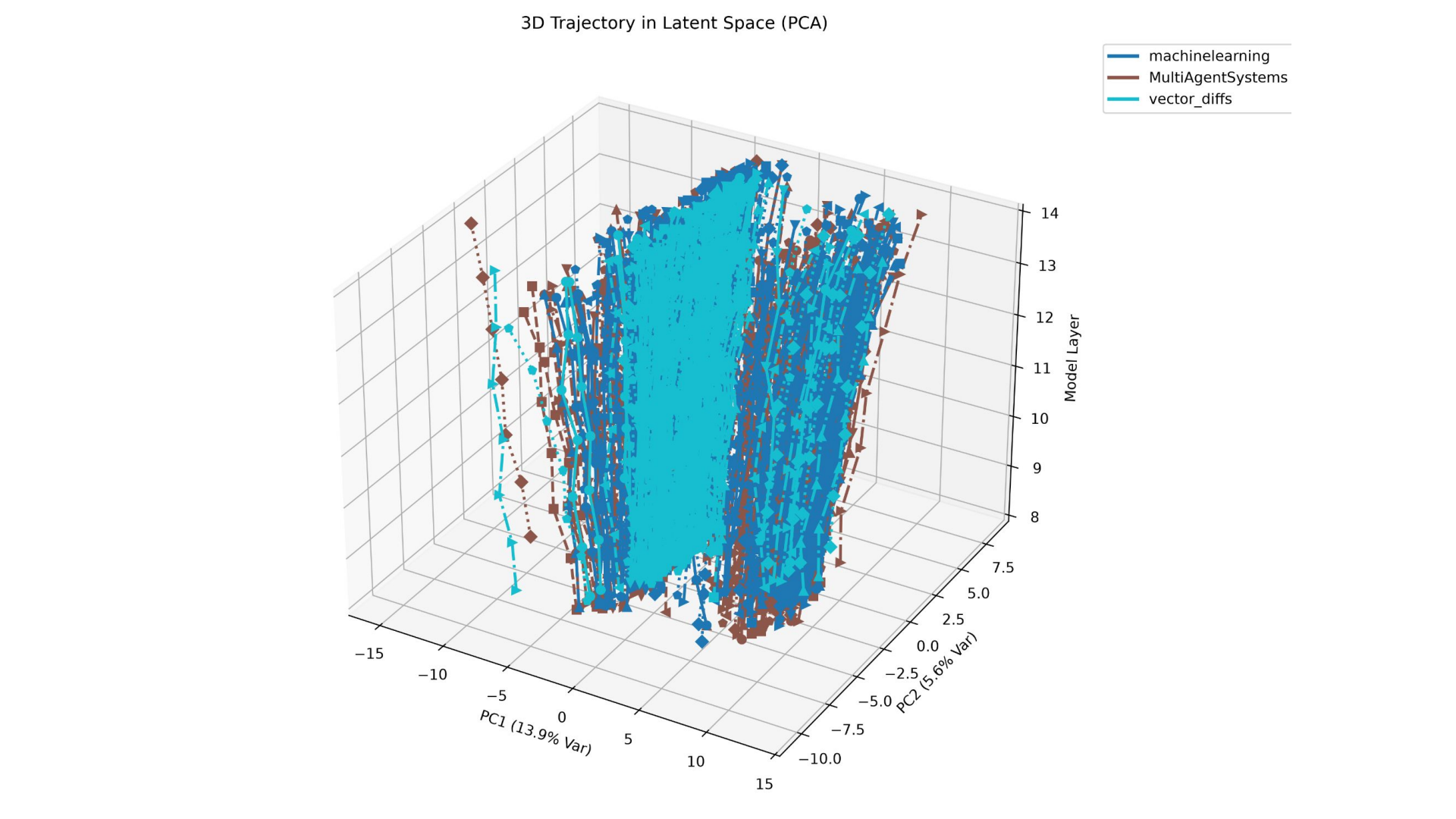
- Using PCA and UMAP in conjunction to reduce dimensions
- Implemented multiple file graphing and better readability
- Experimenting with different degrees of PCA reduction before UMAP


2D Vector Projection Across Layers (PCA)
machinelearning
MultiAgentSystems
vector_diffs
PC2 (5.6% Var)
PC1 (13.9% Var)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−15
−10
−5
0
5
10


2D Vector Projection Across Layers (PCA (50 dims) -> UMAP)
machinelearning
MultiAgentSystems
vector_diffs
UMAP Dimension 2
UMAP Dimension 1
20
15
10
5
0
−5
−10
−15
−10
−5
0
5
10
15
20


2D Vector Projection Across Layers (PCA (50 dims) -> UMAP)
machinelearning
MultiAgentSystems
vector_diffs
UMAP Dimension 2
UMAP Dimension 1
20
15
10
5
0
−5
−10
−15
−10
−5
0
5
10
15
20


3D Trajectory in Latent Space (PCA (50 dims) -> UMAP)
machinelearning
MultiAgentSystems
vector_diffs
Model Layer
14
13
12
11
10
9
8
UMAP Dimension 1
−10
−5
0
5
10
15
20
25
UMAP Dimension 2
20
15
10
5
0
−5
−10
−15


3D Trajectory in Latent Space (PCA)
machinelearning
MultiAgentSystems
vector_diffs
Model Layer
14
13
12
11
10
9
8
PC1 (13.9% Var)
−15
−10
−5
0
5
10
15
PC2 (5.6% Var)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0